

Total No. of Questions : 10]

SEAT No. :

P2

[Total No. of Pages : 3

[5871]-502

B.E. (Civil)

TRANSPORTATION ENGINEERING

(2015 Pattern) (Semester - I)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.No.1 or Q.No. 2, Q.No.3or Q.No.4, Q.No.5 or Q. No.6, Q.No.7 or Q.No.8, Q.No.9 or Q.No.10.
- 2) Figures to the right indicate full marks.
- 3) Use of logarithmic tables, slide rule, Molli's charts, electronics pocket calculator and steam tables is allowed.
- 4) Assume suitable data if necessary.
- 5) Neat diagrams must be drawn wherever necessary.

Q1) a) Roads are the arteries of the nation, on which prosperity of nation spread “comment”? **[5]**

b) Discuss the second twenty year road development plan and its salient features? **[5]**

OR

Q2) The radius of horizontal curve is 400 m, the total pavement width at curve is 7.6 m and the super elevation is 0.07. Design the transition curve length for speed of 100 Kmph. **[10]**

Q3) a) Find the stopping sight distance on a highway at a descending gradient of 1 in 40 for a design speed of 65 Kmph. Assume other data as per IRC recommendations **[5]**

b) Enlist various methods of conducting the Origin and Destination Studies and explain any one in brief. **[5]**

OR

Q4) a) Explain the importance of highway drainage in increasing the serviceability of the road **[5]**

b) Estimate the basic capacity of traffic lane at a speed of 60 Kmph. Assume that all the vehicles are of average length of 6.0 m **[5]**

P.T.O.

- Q5) a)** How are Cutbacks prepared? What are the different types of Cutbacks? State their advantages. [6]
- b) Explain in brief the procedure of conducting 'Ductility Test' on bitumen. [6]
- c) Write a note on types of Bitumen. [5]

OR

- Q6) a)** Explain briefly the Marshall method of bituminous mix design. [6]
- b) How is foamed bitumen produced? state where they can be used. [6]
- c) Write a note on Polymer Modified Bitumen. [5]

- Q7) a)** Calculate the cumulative number of standard axles for a design of new flexible pavement for a two lane undivided carriage way using the following data: [6]

- i) Subgrade soil CBR = 5.0%
- ii) No of heavy vehicles per day in september 2014 = 150
- iii) Design life = 15 years
- iv) Annual rate of increase in the heavy vehicles = 5.0%
- v) Vehicle Damage Factor = 3.5
- vi) Lane Distribution factor = 0.75

The road is proposed to be completed in September, 2019

- b) Explain the concept of ESWL with a neat sketch. [6]
- c) Differentiate between flexible and rigid pavements [5]

OR

- Q8) a)** What is CBR? Calculate the CBR value of a soil sample, if the load sustained by specimen at 2.5 mm and 5.0 mm penetration is recorded as 62.7 kg and 91.2 kg respectively. [6]
- b) Discuss in brief load stresses acting on a rigid pavement. [6]
- c) Define 'Vehicle Damage Factor'. Explain its importance in the design of flexible pavement. [5]

- Q9)** a) Differentiate between Cold Mix and Hot Mix Asphalt Technology. [5]
- b) Explain in brief the construction process of Bituminous Macadam (BM) [6]
- c) Explain the importance of providing prime Coat, tack coat and seal coat in pavements. [5]

OR

- Q10)** a) Write a note on Dry Lean Concrete? [5]
- b) How is the structural evaluation of pavement done using FWD? [6]
- c) Explain how WMM layer is prepared in the field. [5]

